Roots of d-matching polynomials and the spectrum of the universal cover: a closed form at first Betti number one, and a refuted conjecture

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August 2026

Abstract

Hall, Puder and Sawin place the roots of the d-matching polynomial $\mu_{d,G}$ in the interval $[-\rho, \rho]$, ρ the spectral radius of the universal cover T. We conjectured that they lie in $\mathrm{spec}(T)$ itself, a proper subset of that interval whenever T has a spectral gap: no root ever enters a gap. This strengthens their theorem and is the matching-polynomial form of their Question 6.3.

That conjecture is false, and the counterexample is due to Chris Hall [12]. His graph is simple, connected, loopless and bipartite on 41 vertices: five copies of $K_{2,5}$, a pendant leaf on one degree-five vertex of each, and a central vertex joined to the other five. Its matching polynomial is $x^{21}(x^4-11x^2+25)^4(x^2-5)(x^2-11)$, and $\sqrt{5}$ lies in an internal gap of $\mathrm{spec}(T)$, certified by an exact ratio system in the sense of Angel–Friedman–Hoory [13]. We verified his certificate independently, machine-checked its algebra, and show his example is the smallest member of a two-parameter family in which the pendant leaves are essential.

We prove the conjecture for every connected G of first Betti number one and every d, by giving $\mu_{d,G}$ in closed form, and we chart the repair. Minimum degree two fails, bounded maximum degree fails, and 2-connectivity fails. Minimum degree three fails as well, and we give the counterexample: a bipartite graph on 14 vertices, a third the size of Hall's, with $\mu_G = x^4(x^2-3)(x^8-24x^6+171x^4-396x^2+270)$ and $\sqrt{3}$ outside $\mathrm{spec}(T)$. Its divisor is the star $K_{1,3}$, and that is the point: Hall's own violating root $\sqrt{5}$ is the top matching root of $K_{1,5}$, so the engine is a star divisor, whose leaves have degree one in the divisor however large their degree is in G. No hypothesis on the minimum degree of G can reach it. What survives is the connectivity statement, since every counterexample known to us has a separating pair. Everything asserted is machine-checked in Lean 4/Mathlib unless labelled otherwise. The biregular case, which is where the conjecture survives, is the subject of the companion paper [16].

2020 Mathematics Subject Classification. Primary 05C31; Secondary 05C50, 05C70, 68V20.

Keywords. matching polynomial, d-matching polynomial, universal cover, spectral gap, graph covering, interlacing families, Lean 4, formalization.

*Lean 4 sources: `RamaLean/` in the `rama-notes` repository. Correspondence: `vicobonfioli@gmail.com`.

Code and data availability. All computations, the Lean 4/Mathlib sources, and the scripts backing every numerical statement are in the `rama-notes` repository. Each numerical claim names the script that produced it; every named script is executed and its output compared against a recorded baseline by `code/cite_check.py`, and every Lean declaration cited is checked for its axiom base by `code/audit_papers.py`.

Status conventions. Claims here are of three kinds, and the text says which each time rather than leaving it to the reader. A result described as *formalized*, or cited by a Lean declaration name, is machine-checked in Lean 4/Mathlib: the sources are in `RamaLean/` of the accompanying repository, carry no `sorry`, and rest only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. A claim described as *measured*, *verified to* a stated tolerance, or *checked numerically* is computational evidence and is not a proof; the script producing it is named, and every named script is run and its output compared against a recorded baseline (`code/cite_check.py`). A claim called a *conjecture* is unproved, and is called that even where the evidence is strong.

1 Introduction and statement

The expected characteristic polynomial of a random lift is the object at the heart of the interlacing-families method of Marcus–Spielman–Srivastava [11], who used the $r = 2$ (signing) case to prove that bipartite Ramanujan graphs exist in every degree; Hall–Puder–Sawin [2] extended the framework to arbitrary coverings. We evaluate it in closed form for the cycle.

A *permutation r-lift* of a graph G replaces each vertex by r copies and each edge $\{u, v\}$ by the perfect matching $u_i \sim v_{\sigma(i)}$ of a permutation $\sigma \in S_r$ chosen independently and uniformly per edge. For the cycle C_n, gauge-fixing a spanning path reduces the lift to a single permutation $\sigma \in S_r$ on the closing edge, and the lift decomposes into disjoint cycles: each ℓ-cycle of σ contributes a cycle $C_{n\ell}$ (fixed points contribute copies of C_n). Since $\chi_{C_m}(x) = \det(xI - A_{C_m}) = 2T_m(x/2) - 2$, the expected characteristic polynomial is the expected cycle weight

$$\Phi_{n,r}(x) \;=\; \mathbb{E}_{\sigma\in S_r} \prod_{\ell \,\in\, \mathrm{cycles}(\sigma)} \bigl(2T_{n\ell}(x/2) - 2\bigr). \tag{1}$$

Theorem 1. *For all $n \geq 1$, $r \geq 1$, with $Y = T_n(x/2)$,*

$$\Phi_{n,r}(x) \;=\; U_r(Y) - 2U_{r-1}(Y) + U_{r-2}(Y) \;=\; \underbrace{\bigl(2T_n(x/2) - 2\bigr)}_{\chi_{C_n}(x)} \cdot \underbrace{U_{r-1}\bigl(T_n(x/2)\bigr)}_{\Psi_r(x)}.$$

The factored form identifies the quotient Ψ_r, by the theorem of Hall–Puder–Sawin [2] equating the expected new-eigenvalue characteristic polynomial with the d-matching polynomial ($d = r - 1$), as

$$\mu_{d,C_n}(x) \;=\; U_d\bigl(T_n(x/2)\bigr),$$

which is equivalent (through $\mathcal{P}_d(2t) = U_d(t)$, $\mathcal{C}_n(2t) = 2T_n(t)$ and the composition identity $U_{dn+n-1} = U_d(T_n)\, U_{n-1}$) to *Hall's conjecture* $\mathcal{C}_{n,d} = \mathcal{P}_d(\mathcal{C}_n)$, proved combinatorially in [3]. That preprint has remained unrefereed since 2018 and carries no journal reference or DOI as of August 2026. The reason is not mathematical. Chris Hall, who posed the conjecture, has told us (personal communication, August 2026) that Andrew Herring was his first doctoral student; Hall gave him the conjecture, Herring presented it as a problem at a workshop, and the group of [3] solved it there. They submitted the result, met a couple of rejections, and abandoned the attempt for want of motivation rather than for any defect. Hall regards the proof as correct, and so do we. We record the episode because a reader weighing what is established about this identity should know both that the combinatorial proof stands and that the present derivation is, as far as we can tell, the first refereed or machine-checked one. Our route is different and elementary; we make no use of matching-polynomial combinatorics.

Corollary 2. $\sum_{d\geq 0}\mu_{d,C_n}(x)\,z^d = \dfrac{1}{1-2T_n(x/2)\,z+z^2}$, *and* $\sum_{r\geq 0}\Phi_{n,r}(x)\,z^r = \dfrac{(1-z)^2}{1-2T_n(x/2)\,z+z^2}$.

Corollary 3 (real-rootedness)**.** *All roots of $\Phi_{n,r}$ lie in $[-2,2]$: with $x=2\cos\theta$, χ_{C_n} vanishes at $\theta=2\pi m/n$ and $\Psi_r=\sin(rn\theta)/\sin(n\theta)$ vanishes at $\theta=m\pi/(rn)$ with $r\nmid m$ (when $r\mid m$ the denominator vanishes too and $\Psi_r=\pm r\neq 0$); this leaves exactly $n(r-1)$ roots, matching $\deg\Psi_r$.*

Corollary 4 (Fibonacci–Lucas evaluation)**.** $\Phi_{n,r}(3)=(L_{2n}-2)\,F_{2nr}/F_{2n}$, *since* $T_n(3/2)=L_{2n}/2$ *and* $U_{r-1}(L_{2n}/2)=F_{2nr}/F_{2n}$.

Corollary 5 (parity)**.** $\Psi_r(-x)=(-1)^{n(r-1)}\Psi_r(x)$*: the quotient is supported on exponents of a single parity, like a matching polynomial. (For bipartite C_n it is an even polynomial for every r; at $r=2$ it is the classical matching polynomial $2T_n(x/2)$ of C_n, consistent with Godsil–Gutman [1].)*

What this paper does

The closed form above is the starting point rather than the subject. Read as a statement about where the roots of $\mu_{d,G}$ sit, it says that for the cycle they land exactly in $\operatorname{spec}(T)$, the spectrum of the universal cover, and not merely in the interval $[-\rho,\rho]$ that Hall–Puder–Sawin guarantee. Those differ whenever T has a spectral gap, and the question of whether the stronger statement holds in general is the one this paper is about.

The order is as follows. §3 proves the stronger statement for every graph of first Betti number one and every d, by the closed form. §4 states it as a conjecture for all graphs, records what it would contain — the matching-polynomial form of Question 6.3 of [2], and Problem 1 of Song, Fan and Miao — and proves it in two further cases, for subdivisions and for complete bipartite graphs. §5 is the counterexample, due to Chris Hall, which shows the conjecture is false: we verify his certificate, machine-check its algebra, and extend his graph to a two-parameter family in which the pendant leaves turn out to be essential.

The rest of §5 charts the repair. The mechanism behind the counterexample is a *separation*, not low connectivity as the examples first suggest. A local hypothesis, minimum degree three, survives every search we ran against it and is stated as Conjecture 19; §5.6 then refutes it (Proposition 20), on 14 vertices, by a star divisor. Since a star's leaves have degree one in the divisor whatever their degree in G, no lower bound on $\delta(G)$ touches the engine, and the live statement is the connectivity one. The biregular case, which survives the refutation and is where the phenomenon actually lives, is the companion paper [16].

2 Proof

Write $f(\ell) = 2T_\ell(Y) - 2$ with $Y = T_n(x/2)$; by the composition identity $T_{n\ell} = T_\ell \circ T_n$ this is the per-cycle factor in (1).

Step 1 (exponential formula). For any commutative ring R and $f : \mathbb{N} \to R$, the S_r-average $\Phi_r = \mathbb{E}_\sigma \prod_{\ell \in \mathrm{cycles}(\sigma)} f(\ell)$ satisfies

$$r\,\Phi_r = \sum_{k=1}^{r} f(k)\,\Phi_{r-k}, \qquad \Phi_0 = 1, \tag{2}$$

the coefficient form of $\sum_r \Phi_r z^r = \exp\bigl(\sum_\ell f(\ell) z^\ell/\ell\bigr)$. (Proof: group σ by cycle type λ with Cauchy's count $r!/z_\lambda$, mark a part, and re-index; see the Lean file `Paper2ExpFormula.lean` for the fully detailed version.)

Step 2 (Chebyshev generating function). With $f(\ell) = 2T_\ell(Y) - 2$,

$$\sum_{\ell \geq 1} f(\ell)\frac{z^\ell}{\ell} = -\log\bigl(1 - 2Yz + z^2\bigr) + 2\log(1-z) \quad\Longrightarrow\quad \sum_{r \geq 0} \Phi_r z^r = \frac{(1-z)^2}{1 - 2Yz + z^2}.$$

Extracting coefficients against $\sum U_r(Y) z^r = (1 - 2Yz + z^2)^{-1}$ gives $\Phi_r = U_r - 2U_{r-1} + U_{r-2}$, and the three-term recurrence $U_r + U_{r-2} = 2YU_{r-1}$ collapses this to $(2Y-2)\,U_{r-1}(Y)$. $\square$

In the formalization the analytic Step 2 is replaced by an equivalent finite argument: the closed form is shown to satisfy (2) directly (a second-difference induction), and (2) with $\Phi_0 = 1$ determines Φ over a characteristic-zero domain.

3 Graphs of first Betti number one

The proof above uses cycles in exactly one place: gauge-fixing a spanning path leaves a *single* permutation, because C_n has $|E| - |V| + 1 = 1$. The argument therefore applies verbatim to any connected G with $b_1(G) = |E| - |V| + 1 = 1$, a cycle with trees attached, and we record what it gives.

By [2] the expected characteristic polynomial of a random r-lift of *any* graph factors as $\chi_G \cdot \mu_{r-1,G}$; the content below is therefore not that factorization but a closed form for the factor $\mu_{d,G}$ itself. Question 6.5 of [2] asks which parts of the theory of the matching polynomial $\mu_{1,G}$ generalize to $\mu_{d,G}$; the following answers it for $b_1(G) = 1$.

Let C be the unique cycle of G and let $G - V(C)$ be the forest obtained by deleting its vertices. Write μ_H for the ordinary matching polynomial of a graph H, with $\mu_\emptyset = 1$. Define

$$V_0 = 1, \qquad V_1 = \mu_G, \qquad V_d = \mu_G\, V_{d-1} - \mu_{G-V(C)}^2\, V_{d-2}. \tag{3}$$

Theorem 6. *For every connected G with $b_1(G) = 1$ and every $d \geq 0$, $\mu_{d,G} = V_d$.*

Theorem 6 is proved below, through the twisted characteristic polynomial; it is not formalized.

For $G = C_n$ the deleted forest is empty, so $\mu_{G-V(C)} = 1$ and (3) is the recurrence for $U_d(\mu_{C_n}/2) = U_d(T_n(x/2))$: Theorem 1 is the case $\mu_{G-V(C)} = 1$. In general the attached trees enter only through $\mu_{G-V(C)}$; for the triangle with one pendant edge, $\mu_G = x^4 - 4x^2 + 1$ and $\mu_{G-V(C)} = x$.

The proof goes through the twisted characteristic polynomial. Let $e = uv$ be the unique non-tree edge, let $A_G(z)$ be the adjacency matrix of G with the (u, v) entry replaced by z and the (v, u) entry by z^{-1}, and set $F(x, z) = \det(xI - A_G(z))$.

Lemma 7. *$F(x, z) = \mu_G(x) - \mu_{G-V(C)}(x)\,(z + z^{-1})$.*

Proof. By the Sachs coefficient theorem $\det(xI - A)$ expands over elementary subgraphs, that is disjoint unions of edges and cycles. Since a permutation uses each of the two twisted entries at most once and $A_G(z)^{\mathsf{T}} = A_G(z^{-1})$, the determinant is of the form $P(x) + Q(x)(z + z^{-1})$. Replacing z by ± 1 and averaging kills exactly the elementary subgraphs whose cycle traverses e; as G has the single cycle C, those are the subgraphs containing C, so $P = \mu_G$, and the surviving terms contribute $Q = -\mu_{G-V(C)}$, a cycle component carrying the Sachs weight -2. □

Proof of Theorem 6. Gauge-fix a spanning tree: the r-lift is determined by one permutation $\sigma \in S_r$ on e, and its components are indexed by the cycles of σ, the component of an ℓ-cycle being the ℓ-fold cyclic cover G_ℓ. Decomposing the $\mathbb{Z}/\ell$ action into characters gives the classical factorization $\chi_{G_\ell}(x) = \prod_{j=0}^{\ell-1} F(x, \omega^j)$ with $\omega = e^{2\pi i/\ell}$ [9, 10]. With P, Q as in Lemma 7 and $y = -P/(2Q)$,

$$\chi_{G_\ell} = \prod_{j=0}^{\ell-1}\Big(P + 2Q\cos\tfrac{2\pi j}{\ell}\Big) = (-Q)^\ell\big(2T_\ell(y) - 2\big),$$

so by the exponential formula (2), with $w = -Qz$,

$$\sum_{r\geq 0} \Phi_{G,r} z^r = \exp\Big(\sum_{\ell\geq 1}\big(2T_\ell(y) - 2\big)\frac{w^\ell}{\ell}\Big) = \frac{(1-w)^2}{1 - 2yw + w^2},$$

and extracting coefficients as in Section 2 gives $\Phi_{G,r} = \chi_G \cdot (-Q)^{r-1} U_{r-1}(y)$. Comparing with $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ [2] yields $\mu_{d,G} = (-Q)^d U_d(y)$, which satisfies (3). □

Since $U_d(y) = 2^d \prod_{k=1}^d (y - \cos\frac{k\pi}{d+1})$, the theorem can be restated as a product, $\mu_{d,G}(x) = \prod_{k=1}^d F\big(x, e^{ik\pi/(d+1)}\big)$, whence:

Corollary 8. *For $b_1(G) = 1$, $\mu_{d,G}$ is a product of d real-rooted polynomials, and its roots are exactly the eigenvalues of the Hermitian matrices $A_G\big(e^{ik\pi/(d+1)}\big)$, $1 \leq k \leq d$.*

This is the real-rootedness theorem of [2] for this family, with the roots identified. The localization itself, that a vanishing V_d forces $|\mu_G| \leq 2|\mu_{G-V(C)}|$, so that a root can never sit in a spectral gap, is machine-checked in Lean 4 as `cV_root_mem_band`, resting on `cU_ne_zero_of_one_le_abs`: the Chebyshev polynomial U_d has no root of absolute value at least one.

The roots can be located much more precisely than that. Since $b_1(G) = 1$ the group $\pi_1(G)$ is $\mathbb{Z}$, so the universal cover T of G *is* the $\mathbb{Z}$-cover, and T is a periodic decorated chain whose adjacency operator decomposes by Floquet–Bloch into the fibres $A_G(e^{i\theta})$. Hence

$$\operatorname{spec}(T) = \bigcup_\theta \operatorname{spec} A_G(e^{i\theta}) = \Big\{x :\ \exists\, s \in [-1, 1],\ \mu_G(x) = 2s\,\mu_{G-V(C)}(x)\Big\},$$

a finite union of bands, together with a flat band at each common zero of μ_G and $\mu_{G-V(C)}$, where $F(x, \cdot)$ vanishes identically.

Corollary 9. *For $b_1(G) = 1$ the roots of $\mu_{d,G}$ are exactly the solutions of $\mu_G(x) = 2\cos\big(\frac{k\pi}{d+1}\big)\mu_{G-V(C)}(x)$ for $1 \le k \le d$. In particular every root of every $\mu_{d,G}$ lies in* $\mathrm{spec}(T)$*, and as $d \to \infty$ the roots equidistribute on* $\mathrm{spec}(T)$ *with respect to its density of states.*

Proof. Immediate from Theorem 6 in the form $\mu_{d,G} = \prod_k F(x, e^{ik\pi/(d+1)})$ and $F(x, e^{i\theta}) = \mu_G - 2\cos\theta\, \mu_{G-V(C)}$. The equidistribution is the arcsine law for $\cos\big(k\pi/(d+1)\big)$ transported through the fibre map. □

Hall–Puder–Sawin place the roots of $\mu_{d,G}$ in the Ramanujan *interval* $[-\rho, \rho]$, ρ being the spectral radius of T. Corollary 9 places them in $\mathrm{spec}(T)$, which is a union of bands and is a proper subset of that interval whenever T has a spectral gap: no root of any $\mu_{d,G}$ can ever enter a gap. The two statements coincide exactly when the band structure is trivial, and that is precisely the case of a cycle, where $\mu_{G-V(C)} = 1$ and $\mathrm{spec}(T) = [-2, 2]$ is the whole interval. Attaching even a single pendant edge opens gaps, and they are not small (computed band structures, proportion of $[-\rho, \rho]$ that is a gap):

G	ρ	gap proportion
C_n	2	0%
triangle + pendant	2.170	36.5%
C_4 + pendant	2.136	45.9%
triangle + P_2	2.214	51.3%
$C_4 + P_3$	2.189	54.1%
triangle + $K_{1,3}$	2.514	65.9%

So the cycle is not merely the easiest case of Theorem 6: it is the degenerate one, the unique member of the family whose spectrum has no gaps and for which the Ramanujan interval is attained. Any cycle with a tree attached exhibits a band structure that no cycle can display.

The boundary is sharp. For $b_1(G) = b$ the lift is determined by b permutations, that is by an action of the free group F_b on $[r]$, whose orbits are the components of the lift. The exponential formula still applies, in the form $N_r = \sum_\ell \binom{r-1}{\ell-1} C_\ell N_{r-\ell}$ with $N_r = (r!)^b \Phi_{G,r}$ and C_ℓ the sum of χ over the transitive b-tuples on $[\ell]$; we have verified this for $b \le 3$. What fails is the atom. For $b = 1$ every transitive tuple gives the *same* cover G_ℓ, so $C_\ell = (\ell-1)!\,\chi_{G_\ell}$: one graph per ℓ, with χ_{G_ℓ} Chebyshev in ℓ. For $b \ge 2$ the transitive tuples number $(\ell-1)!$ times the count of index-ℓ subgroups of F_b and spread over many non-isomorphic covers with distinct spectra, so C_ℓ carries no such structure. Accordingly Theorem 6 is not merely unproved for $b \ge 2$: it is false there already at $d = 1$, as we checked for the theta graph, the bowtie, K_4, $K_{2,3}$ and two triangles joined by an edge.

Theorem 6 was verified by exact integer computation against the definition of $\mu_{d,G}$ as an average of matching polynomials over all d-covers, and against brute-forced lift averages, on thirteen named unicyclic graphs for $r \le 6$ and on a random sample of thirty-four for $r \le 5$; Lemma 7 was checked on eleven.

4 A conjecture for all graphs

Corollary 9 is proved by Floquet theory, which is available only because $\pi_1(G) = \mathbb{Z}$. Its *statement*, however, makes sense for every graph, and we know one case of it: at $b_1(G) = 1$

it is Corollary 9, for every d. We therefore propose:

Conjecture 10. *For every finite graph G and every $d \geq 1$, all roots of $\mu_{d,G}$ lie in* $\mathrm{spec}(T)$*, the spectrum of the universal cover of G.*

Theorem 11 (Hall [12])**.** *Conjecture 10 is false, already at $d = 1$ and already for simple connected loopless bipartite graphs.*

The counterexample and its certificate are entirely due to Chris Hall. We record it here because it settles the central question of this paper, and because everything else in the note has to be read in its light. §5 states his construction, our independent verification of it, and what we can add.

The biregular case of Conjecture 10 at $d = 1$ is not new. It is Problem 1 of Song, Fan and Miao [5], posed in 2023: they ask whether the matching polynomial of the incidence graph B_H of a (d, r)-regular hypergraph has no root μ with $0 < |\mu| < |\sqrt{d-1} - \sqrt{r-1}|$, and every (d, r)-biregular bipartite graph is such an incidence graph. Their Theorem 1.8 attaches a consequence: an affirmative answer would give every (d, r)-regular hypergraph a left-sided Ramanujan 2-covering. The only bound in that direction we know of is their Lemma 5.5, which goes the other way, $0 < \mu_\tau(B_H) \leq \max\{\sqrt{d}, \sqrt{r}\}$. We are not aware of any proof, partial result or counterexample. Conjecture 10 extends their problem in two directions, to arbitrary graphs and to all d.

Question 6.3 of [2] asks the corresponding question for graph eigenvalues, and leaves it open: it distinguishes graphs whose nontrivial spectrum lies in the *interval* $[-\rho, \rho]$ from those whose nontrivial spectrum lies in $\mathrm{spec}(T)$, observes that the two notions separate exactly when T has a gap, and gives the biregular tree as the example. Conjecture 10 is the matching-polynomial form of that distinction.

The conjecture strengthens the root-localization theorem of [2], which places the roots in the interval $[-\rho, \rho]$ with ρ the spectral radius of T; the conjecture places them in $\mathrm{spec}(T)$ itself. The two agree whenever T has no spectral gap, in particular for regular G, where T is a regular tree and $\mathrm{spec}(T) = [-2\sqrt{q}, 2\sqrt{q}]$ is an interval. They differ exactly when T has a gap, and the smallest examples are biregular: for $K_{2,3}$ the universal cover is the $(3, 2)$-biregular tree, with spectrum $\{0\} \cup \pm[\sqrt{2}-1, \sqrt{2}+1]$ and hence a gap $(0, \sqrt{2}-1)$ inside the Ramanujan interval. We have tested the conjecture on the biregular family, where the universal cover is an (a, b)-biregular tree with spectrum $\{0\} \cup \pm[\,|s-t|, s+t\,]$, $s = \sqrt{a-1}$, $t = \sqrt{b-1}$, so the gap is $(0, |s-t|)$. In each case $\mu_{d,G}$ was computed by averaging matching polynomials over all d-covers. No root ever lies in the gap; we also record the smallest nonzero $|\mathrm{root}|$ as a multiple of the gap edge, which measures how close the bound comes to being attained.

G	b_1	type	gap edge	d	min $\lvert\mathrm{root}\rvert$/gap edge
$K_{2,3}$	2	$(3, 2)$	0.4142	$1, 2, 3, 4$	2.72, 2.19, 1.94, 1.78
$K_{2,4}$	3	$(4, 2)$	0.7321	$1, 2, 3$	1.93, 1.65, 1.52
$K_{2,5}$	4	$(5, 2)$	1.0000	$1, 2, 3$	1.66, 1.46, 1.37
$K_{3,4}$	6	$(4, 3)$	0.3178	$1, 2$	3.04, 2.31
$K_{3,5}$	8	$(5, 3)$	0.5858	$1, 2$	2.10, 1.73
subdivision of K_4	3	$(3, 2)$	0.4142	$1, 2$	1.97, 1.67

All of these have $b_1 \geq 2$, so none is covered by Corollary 9. It is worth recording that even $d = 1$ is not formal. There $\mu_{1,G}$ is the ordinary matching polynomial, whose roots are eigenvalues of the path tree of G (Godsil), and the path tree embeds in T; but eigenvalues of a finite subtree of T need not lie in $\mathrm{spec}(T)$ when T has a gap. For the $(3,2)$-biregular tree, whose gap is $(0, \sqrt{2}-1)$, the path P_8 is a subtree with eigenvalue $2\cos(4\pi/9) = 0.3473$ inside that gap. So the $d=1$ case does not follow from the path-tree argument. It is nevertheless a theorem for an infinite family.

4.1 Subdivisions: the case of minimum degree two

Theorem 12. *Let H be D-regular, $D \geq 2$, and let $S(H)$ be its subdivision, so that the universal cover of $S(H)$ is the $(D,2)$-biregular tree with spectral gap $\left(0, \sqrt{D-1}-1\right)$. Then every nonzero root of $\mu_{S(H)}$ has absolute value at least $\sqrt{D-1}-1$; that is, Conjecture 10 holds for $S(H)$ at $d=1$. The constant is the spectral gap of the universal cover; we do not claim it is attained, and by the strictness of Heilmann–Lieb on finite graphs it is not.*

Theorem 12 settles the case $\min(d,r)=2$ of Problem 1 of [5]. Their uniformity condition allows $r \geq 2$, and they observe that when H is a simple graph the incidence graph B_H is exactly the subdivision $S(H)$; so their question at $r=2$ asks precisely whether $\mu_{S(H)}$ has no root μ with $0 < |\mu| < |\sqrt{D-1}-1|$. By their Theorem 1.8 this gives every D-regular graph, viewed as a $(D,2)$-regular hypergraph, a left-sided Ramanujan 2-covering. The case $d=2$ follows as well, since B_H is a bipartite graph and μ_{B_H} does not distinguish its two sides: a $(2,r)$-regular hypergraph has for incidence graph the subdivision of an r-regular graph, and the quantity $|\sqrt{d-1}-\sqrt{r-1}|$ is symmetric in d and r. The case $\min(d,r) \geq 3$ of their problem remains open in general, as does Conjecture 10 for $\min(a,b) \geq 3$; Theorem 13 below settles the complete bipartite graphs inside that range, $K_{3,4}$ among them.

Proof. By the subdivision identity $\mu_{S(H)}(x) = x^{|E|-|V|}\,\mu_H(x^2-D)$ [4], the nonzero roots of $\mu_{S(H)}$ are the x with $x^2 = D+\theta$ for θ a root of μ_H. Heilmann–Lieb gives $\theta \geq -2\sqrt{D-1}$, so with $s=\sqrt{D-1}$ we get $x^2 \geq s^2+1-2s = (s-1)^2$, whence $|x| \geq s-1$. Sharpness: for cubic H of large girth $\theta_{\min} \to -2\sqrt{2}$, so the bound is approached. □

The inequality step is machine-checked in Lean 4 as `subdivision_gap`. The theorem is exactly Heilmann–Lieb for H transported through the subdivision identity, and the numbers match: for $H = K_4$, $\mu_{K_4} = x^4 - 6x^2 + 3$ gives smallest positive root $\sqrt{3-\sqrt{3+\sqrt{6}}} = 0.8158$, against the gap edge 0.4142; the subdivisions of $K_{3,3}$, the prism, Q_3, the Wagner, Petersen, Heawood and Möbius–Kantor graphs give $0.701, 0.714, 0.650, 0.647, 0.607, 0.563, 0.551$, decreasing toward the gap edge with the girth and never reaching it.

For $K_{2,3}$ the computation was pushed to $d=6$ using $\mu_{d,G} = \mathbb{E}[\chi_H]/\chi_G$ over $(d+1)$-covers, which is polynomial time per cover, together with the reduction of the first free permutation to one representative per cycle type; the ratios continue 1.68 at $d=5$ and no root in the gap at $d=6$. That last case is a caution worth recording: in floating point it appeared to produce roots at ± 0.055, well inside the gap, which an exact integer recomputation showed to be an artefact of evaluating a degree-30 polynomial built from 35×35 characteristic polynomials in double precision.

The last column decreases monotonically in d in every case, which is what the conjecture predicts: by Corollary 9 in the $b_1 = 1$ case the roots equidistribute on $\mathrm{spec}(T)$, so they should approach the gap edge without crossing it. The evidence is therefore not merely that the bound holds but that it is approached, though no computation here reaches the regime where the ratio is close to 1.

4.2 Complete bipartite graphs

The smallest graph in the open range $\min(a, b) \geq 3$ is $K_{3,4}$, which is $(4, 3)$-biregular. The complete bipartite family is settled outright, by an exact computation rather than by transport from a regular graph.

Theorem 13. *Let $2 \leq p \leq q$. Every root of $\mu_{K_{p,q}}$ lies in*

$$\{0\} \cup \pm\big[\sqrt{q-1} - \sqrt{p-1},\ \sqrt{q-1} + \sqrt{p-1}\big],$$

the spectrum of the (q, p)-biregular tree. So Conjecture 10 holds for every $K_{p,q}$ at $d = 1$.

Proof. Since $m_k(K_{p,q}) = \binom{p}{k}\binom{q}{k}k!$, comparison of coefficients with $L_p^{(\alpha)}(t) = \sum_k (-1)^k \binom{p+\alpha}{p-k} t^k/k!$ at $\alpha = q - p$ gives

$$\mu_{K_{p,q}}(x) = x^{\,q-p}(-1)^p\, p!\, L_p^{(q-p)}(x^2),$$

so the nonzero roots are the square roots of the zeros of a Laguerre polynomial. Those zeros are the eigenvalues of the $p \times p$ symmetric tridiagonal Jacobi matrix J with $J_{kk} = 2k+\alpha+1$ and $J_{k,k+1} = J_{k+1,k} = \sqrt{(k+1)(k+\alpha+1)}$, $0 \leq k \leq p-1$.

Put $c = p+q-2$ and $g(u) = \sqrt{u(u+\alpha)} - u$, and apply Gershgorin to $J - cI$. The shifted diagonal in row k is $2k-2p+3$, which is at most -1 for $k \leq p-2$, so the radius of an interior row is

$$(2p-3-2k) + \sqrt{k(k+\alpha)} + \sqrt{(k+1)(k+1+\alpha)} \;=\; 2p-2+g(k)+g(k+1),$$

the index cancelling exactly. The step inequality

$$\sqrt{u(u+\alpha)} + 1 \;\leq\; \sqrt{(u+1)(u+1+\alpha)} \tag{4}$$

reduces, on squaring, to $\sqrt{u(u+\alpha)} \leq u + \alpha/2$, which is $\alpha^2 \geq 0$. It makes g nondecreasing along the integers, so the radius is largest at $k = p-2$, where it is $1 + \sqrt{(p-2)(p-2+\alpha)} + \sqrt{(p-1)(q-1)}$ and (4) at $u = p-2$ bounds it by $2\sqrt{(p-1)(q-1)}$. The last row has shifted diagonal 1 and the single off-diagonal entry $\sqrt{(p-1)(q-1)}$, so its radius lies within the same bound as soon as $(p-1)(q-1) \geq 1$, which holds for $p \geq 2$. Hence every eigenvalue t of J satisfies $|t - (p+q-2)| \leq 2\sqrt{(p-1)(q-1)}$; since $\big(\sqrt{q-1} \pm \sqrt{p-1}\big)^2 = (p+q-2) \pm 2\sqrt{(p-1)(q-1)}$, that disc is exactly the stated band for $x^2 = t$. The factor x^{q-p} contributes the root 0, which lies in the tree spectrum. □

The bound is sharp: at $p = q$ every interior Gershgorin row attains it, and the conclusion is then the Heilmann–Lieb interval for the p-regular $K_{p,p}$. It does need $p \geq 2$: at $p = 1$ the only zero is q while the band degenerates to $\{q-1\}$, the $(q, 1)$-"biregular tree" being the finite star $K_{1,q}$. For $K_{3,4}$ the Laguerre factor is $t^3 - 12t^2 + 36t - 24$, with zeros $0.935,\ 3.30,\ 7.76$ against the band $[0.101,\ 9.899]$.

Inequality (4), the two row bounds, the endpoint identity and the Gershgorin step are machine-checked in Lean 4 as `LaguerreBand.sqrt_step`, `rowRadius_le`, `lastRow_le`, `band_endpoints` and `laguerre_band`. The coefficient identification and the passage to the Jacobi matrix are classical and remain by hand.

4.3 Three further reductions

The following arose from a conversation Chris Hall conducted with a language model and shared with us in August 2026. We have checked them, and record them here with their status, which differs sharply between the three. None is used elsewhere in this paper.

Localization in the maximal abelian cover. Put a circle variable on each of the $b_1(G)$ cotree edges and let $A_G(z)$ be the resulting magnetic adjacency matrix. Haar integration over $\mathbb{T}^{b_1}$ annihilates every permutation cycle of length at least three, since a simple cycle must use a cotree edge and therefore carries a nontrivial character; fixed points and transpositions survive, and those are exactly the matchings. Hence

$$\mu_G(x) = \int_{\mathbb{T}^{b_1}} \det\bigl(xI - A_G(z)\bigr)\, dz.$$

For $|z| = 1$ the matrix $A_G(z)$ is Hermitian, so the integrand is real for real x; and a continuous real function on a connected space whose average vanishes must vanish somewhere. Therefore every matching root is an eigenvalue of some $A_G(z)$, that is $\mathrm{Zeros}(\mu_G) \subseteq \mathrm{spec}(G^{\mathrm{ab}})$, and Conjecture 10 at $d = 1$ becomes

$$\mathrm{Zeros}(\mu_G) \cap \bigl(\mathrm{spec}(G^{\mathrm{ab}}) \setminus \mathrm{spec}(T)\bigr) = \emptyset.$$

The realness of the integrand, the vanishing-average argument and the reduction are machine-checked (`AbelianCover.det_sub_smul_one_im`, `exists_zero_of_integral_zero`, `root_mem_of_average`, `reduction`). The cancellation over cycle types, left by hand above, reduces to one statement: a sum of nonzero vectors with pairwise disjoint supports is nonzero. The cycles of a permutation are vertex-disjoint, hence use disjoint edge sets, hence have homology classes of disjoint support, and each class is nonzero because a cycle must use a cotree edge. That lemma is `TorusGodsilGutman.sum_ne_zero_of_disjoint_support`, and the sign and averaging steps are re-proved in the same file in the compact form `sign_const_of_nonvanishing`, `nonvanishing_average_ne_zero`, `mu_ne_zero_of_nonvanishing`, the last being the contrapositive actually used. The two graph-theoretic inputs it consumes are also machine-checked. That every cycle uses at least one cotree edge is `CotreeCycle.exists_cotree_edge`: if all its edges lay in the tree, the walk would transfer to a cycle there, which acyclicity forbids, and no relation between the tree and G is needed beyond acyclicity. That vertex-disjoint cycles use disjoint edge sets is `disjoint_edges_of_disjoint_support`. What remains informal is only the identification of the monomial attached to a permutation with the sum of the homology classes of its cycles, which is bookkeeping about how the magnetic matrix is set up rather than a mathematical step. The identity is checked numerically to $7 \cdot 10^{-15}$ on seven graphs with b_1 up to four (`code/torus_gg.py`).

It is worth being precise about what the connectedness of $\mathbb{T}^{b_1}$ contributes, since Godsil–Gutman average over the $2^{|E|}$ signings and obtain the same identity. A nonvanishing continuous function on a finite discrete set may take both signs, so no conclusion about the average follows; on a connected space it may not. The inclusion $\mathrm{Zeros}(\mu_G) \subseteq \mathrm{spec}(G^{\mathrm{ab}})$ is a consequence of the topology of the parameter space, not of the averaging.

The localization is useful in the negative direction too: any counterexample must live in $\mathrm{spec}(G^{\mathrm{ab}}) \setminus \mathrm{spec}(T)$.

Two 2025 papers settle the companion question for the *point* spectrum, in the opposite direction. Li, Magee, Sabri and Thomas [14] characterise the eigenvalues of G^{ab} as the reals that are roots of $\mu_{G\setminus\Gamma}$ for every 2-regular subgraph Γ, which at $\Gamma = \emptyset$ gives $\operatorname{pointspec}(G^{\mathrm{ab}}) \subseteq \operatorname{Zeros}(\mu_G)$; Spier [15] proves that G^{uni} and G^{ab} have the same eigenvalues. Those concern flat bands, so neither contains nor is contained in the inclusion above, which concerns the full spectrum. But Spier's theorem says the two covers are indistinguishable at the level of point spectrum, and the residue $\operatorname{spec}(G^{\mathrm{ab}}) \setminus \operatorname{spec}(T)$ is exactly where they become distinguishable, so that is where a proof of Conjecture 10 has to act.

Feedback vertex number one. Suppose some vertex v has $F = G - v$ a forest, so that every cycle of G passes through v; the first Betti number is then unrestricted, and flowers, theta graphs and every $K_{2,q}$ qualify. Writing the universal cover's adjacency operator as a matrix over $C^*_r(F_{b_1})$ and eliminating the preimage of F, the Schur complement $S_v(x)$ onto the fibre over v is a *finite* sum of group elements, so it lies in the group algebra inside $C^*_r(F_{b_1})$ rather than merely in the von Neumann algebra $L(F_{b_1})$. That is the load-bearing point, since $L(F_{b_1})$ is a II_1 factor full of projections while $C^*_r(F_{b_1})$ is projectionless by Pimsner–Voiculescu. A self-adjoint invertible element of a projectionless C^*-algebra is definite, so its canonical trace cannot vanish. And that trace is

$$\tau\bigl(S_v(x)\bigr) = x - \sum_{p\sim v} \frac{\mu_{F-p}(x)}{\mu_F(x)} = \frac{\mu_G(x)}{\mu_{G-v}(x)},$$

the first equality because returning to the same lift of v forces a walk to enter and leave a lifted component of F at the same attachment point (two distinct attachment points of one component sit under distinct lifts of v, else the universal cover would contain a cycle), and the second by the matching recursion at v together with the cofactor formula for a forest resolvent. A root of μ_G that is not a root of μ_{G-v} therefore makes $\tau(S_v)$ vanish, forcing S_v to be non-invertible, hence the root lies in $\operatorname{spec}(T)$.

We machine-check the pieces that can be: the combinatorial core (`FeedbackVertex.no_bypass`), the resolvent diagonal (`resolvent_diag`), the algebraic collapse of the trace formula (`matching_ratio`), and the trace step (`trace_ne_zero_of_definite`). Pimsner–Voiculescu is not in Mathlib, so `feedback_one_of` carries it as an explicit hypothesis in the shape the argument consumes. We therefore state this as established modulo that classical input, and not as machine-checked end to end.

Two closure operations. Attaching a fixed rooted tree R at every vertex sends $\mu_{d,H}$ to $\beta^N \mu_{d,H}(\alpha/\beta)$ with $\alpha = \mu_R$, $\beta = \mu_{R-o}$ and $N = d\,|V(H)|$, because an uncovered vertex leaves its whole copy of R free while a covered one leaves $R - o$; the same ratio α/β is what the Schur complement of the universal cover produces, so the conjecture is inherited. Starting from a regular graph, where the conjecture is immediate, this yields graphs of arbitrarily large first Betti number whose universal covers have genuine gaps. Independently, the subdivision identity survives averaging over d-covers, since suppressing the degree-two fibres of a cover of $S(H)$ gives a cover of H and a product of independent uniform permutations is uniform; that upgrades Theorem 12 from $d = 1$ to every d, and with it the whole $(D, 2)$-biregular family. The substitution identity and the root transfer are machine-checked (`TreeSubstitution.homogenize`, `star_ratio`, `subdivision_root`, `subdivision_abs`); the covering combinatorics, which is the same fibre bookkeeping as `CoverCounts`, is by hand. Neither operation touches the irreducible core: both reduce the spectral problem to a *scalar* transfer function, and the difficulty begins where elimination leaves a matrix-valued one.

4.4 The biregular case is one interval, and we have one end of it

Let G be (a, b)-biregular bipartite, $s = \sqrt{a-1}$, $t = \sqrt{b-1}$. Since G is bipartite, $\mu_G(x) = x^\varepsilon f(x^2)$ for a polynomial f and $\varepsilon \in \{0, 1, 2, \dots\}$. Put

$$c = (a-1) + (b-1), \qquad m = (a-1)(b-1), \qquad g(z) = f(z+c).$$

The universal cover has spectrum $\{0\} \cup \pm[\,|s-t|, s+t\,]$, and $(s \pm t)^2 = c \pm 2\sqrt{m}$, so

$$x \in \operatorname{spec}(T) \setminus \{0\} \iff x^2 - c \in \left[-2\sqrt{m},\ 2\sqrt{m}\right].$$

The change of variable therefore turns Conjecture 10, for biregular G, into a single clean statement:

$$\text{every root of } g \text{ lies in } \left[-2\sqrt{m}, 2\sqrt{m}\right], \tag{5}$$

which is a Heilmann–Lieb interval of effective degree $m + 1$.

It is worth recording what this says about the matching generating polynomial $M_G(w) = \sum_k m_k w^k$, since that, rather than μ_G, is the object of [6]. From $f_G(y) = y^p M_G(-1/y)$ the roots correspond by $w = -1/y$, so a lower bound on the least positive root of f_G is an upper bound on the modulus of the roots of M_G. Writing $Q_G(z) = M_G(-z) = \sum_k (-1)^k m_k z^k$, the reciprocal-and-reflect of f_G:

$$\text{Conjecture 10 at } d = 1 \iff \text{every root of } Q_G \text{ lies in } \Big(0,\ (s-t)^{-2}\Big]. \tag{6}$$

Heilmann–Lieb, in the same variable, says every root of Q_G is at least $\left(2\sqrt{\Delta - 1}\right)^{-2}$ in modulus. The two statements bound the roots of one polynomial from the two sides, so (5) and (6) are the same dichotomy seen in reciprocal coordinates. This is a restatement and nothing more, but it is the coordinate in which the classical machinery is written, and Q_G is the polynomial whose deletion recursion $Q_G = Q_{G-v} - z \sum_{u\sim v} Q_{G-v-u}$ is formalized in the accompanying Lean development. The upper end of (5) is known: the path-tree argument of Godsil bounds the roots of μ_G by the spectral radius of the path tree, which for an (a, b)-biregular graph with $a, b \geq 2$ is at most $s + t$ [7, 8]. It is worth noting that [6] proved the corresponding statement only in the regular case; the irregular case is due to Godsil and to [11]. The lower end is the conjecture. *They are the two ends of one interval*, so what is missing is exactly half of a two-sided Heilmann–Lieb theorem for biregular bipartite graphs.

5 Hall's counterexample

Everything in this section except §5.3 is due to Chris Hall, who constructed the graph, found the certificate, and wrote the verification note [12]. Our contribution is the independent check, the formalization of the algebra, and the family in §5.3.

5.1 The graph and its matching polynomial

Let G consist of a central vertex c; for $1 \le i \le 5$ a copy of $K_{2,5}$ with degree-five vertices v_i, w_i and middle vertices $u_{i,1}, \ldots, u_{i,5}$; a pendant leaf ℓ_i at each w_i; and the edges cv_i. It is simple, connected, loopless and bipartite, with 41 vertices, 60 edges and $b_1(G) = 20$.

Writing H for one branch after deleting c, the vertex-deletion recurrence gives $\mu_{H-v} = x^5(x^2 - 6)$ and $\mu_H = x^4(x^4 - 11x^2 + 25)$, and then

$$\mu_G(x) = \mu_H(x)^4\big(x\mu_H(x) - 5\mu_{H-v}(x)\big) = x^{21}(x^4 - 11x^2 + 25)^4(x^2 - 5)(x^2 - 11), \qquad (7)$$

so $\sqrt{5}$ is a simple root. Identity (7), the simplicity of the root, and the value of the cofactor are machine-checked in `RamaLean/HallCounterexample.lean` as an identity in $\mathbb{Z}[X]$ (`muG_eq`, `muG_root_sqrt5`, `muG_root_simple`).

5.2 The spectral exclusion

Angel, Friedman and Hoory [13] characterise invertibility of $A_T - \lambda I$ by the existence of a finite nonzero *ratio system*: an assignment r_e to directed edges with $\lambda = 1/r_e + \sum_{e\to f} r_f$, whose decay matrix $\mathcal{R}_{e,f} = |r_f|^2$ on non-backtracking pairs has spectral radius below one. Hall exhibits such a system at $\lambda = \sqrt{5}$ with eight edge types and ratios in $\mathbb{Q}(\sqrt{5}, \sqrt{41})$. Its decay matrix has 120 states; the ten states at the leaves are transient, and the recurrent block of 110 states reduces by symmetry to a 6×6 orbit quotient K for which

$$x = (20, 121, 47, 33, 28, 16)^{\mathsf{T}} \quad \text{satisfies} \quad Kx < x$$

componentwise, so $\rho(K) < 1$ by Perron–Frobenius. Hence $\sqrt{5} \notin \operatorname{spec}(A_T)$, and with (7) this proves Theorem 11. Since G is bipartite the same holds at $-\sqrt{5}$, and Hall–Puder–Sawin place $\pm\sqrt{5}$ inside $[-\rho(T), \rho(T)]$, so these are roots in a genuine internal gap.

The eight ratio identities, the elementary bounds $32/5 < \sqrt{41} < 13/2$, the strict positivity of all six components of $x - Kx$, and a Collatz–Wielandt lemma (a nonnegative matrix with a positive strictly subinvariant vector has spectral radius below one) are machine-checked in `RamaLean/HallCounterexample.lean`. The argument common to all three counterexamples of this paper is isolated once in `RamaLean/RatioCertificate.lean`: a ratio system with its decay matrix (`decay`, `decay_nonneg`), the Collatz–Wielandt step in the strict form the certificate needs (`decay_lt_one`), and the assembly with Angel–Friedman–Hoory as an explicit hypothesis (`spec_excluded`). The three instances differ only in the numbers supplied. A schema rather than closed-form algebra is the right level here: Hall's ratios lie in $\mathbb{Q}(\sqrt{5}, \sqrt{41})$, but that is a feature of his graph, and for the 31-vertex one the ratios admit no minimal polynomial of small degree and moderate height. The theorem of [13] is not in Mathlib and enters as an explicit hypothesis, in the shape it is consumed.

We verified the certificate independently, reconstructing every object from the graph rather than copying the displayed matrices: the matching numbers $(1, 60, 1585, 24260, 238105, 1564976, 6973855, 20805500, 39784375, 44062500, 21484375)$; the ratio equations on all 120 directed edges; the strongly connected components of the decay digraph; the orbit quotient rebuilt from follower counts; and the decay rate by two independent routes, the componentwise certificate and the characteristic polynomial $t^6 - \frac{201+19s}{800}t^2 + \frac{21s-241}{250}$ with $s = \sqrt{41}$. That reconstruction is `code/hall_certificate.py`, which rebuilds each object from the construction and copies no displayed matrix: the matching numbers agree term for term and $\mu_G(\sqrt{5})$ vanishes to 10^{-45}; the ratio system reduces to eight unknowns by symmetry and is solved exactly in $\mathbb{Q}(\sqrt{5}, \sqrt{41})$, its solution satisfying all 120 equations to 4×10^{-16}; the ten transient states are exactly the leaf-incident directed edges and the recurrent block has 110; and the six components of $x - Kx$ reproduce the exact expressions carried by `RamaLean/HallCounterexample.lean` to 3×10^{-14}, with $\rho(K) = 0.9636233789$. Three of the eight ratios are negative, which is why a positive cavity iteration does not find the system. As a physical check, the non-backtracking cavity Green function at $\sqrt{5} + i\eta$ converges to Hall's real ratios as $\eta \downarrow 0$, with density of states vanishing linearly in η.

5.3 A family, and the role of the leaves

Hall's graph is the case $p = q = 5$ of the family $G(p, q)$ with p branches, each a copy of $K_{2,q}$ carrying a pendant leaf. The branch arithmetic is uniform: the last factor of $\mu_{G(p,q)}$ is

$$x^q\bigl(x^4 - (2q + 1 + p)x^2 + q^2 + p(q + 1)\bigr).$$

Whether its smaller root lies in a gap of $\mathrm{spec}(T)$ is a separate question from whether that root is rational, and the two are independent: over $2 \le p, q \le 7$ the discriminant is a perfect square in six cases, while nine cases give a root inside a gap, and the two sets overlap in only three. The counterexamples are

$$(p, q) \in \{(7, 3), (6, 4), (7, 4), (5, 5), (6, 5), (7, 5), (6, 6), (7, 6), (7, 7)\},$$

with $41, 43, 43, 49, 50, 55, 57, 64, 71$ vertices; Hall's $(5, 5)$ is the smallest. The least p that works is $7, 6, 5, 6, 7$ for $q = 3, 4, 5, 6, 7$, so the phenomenon is cheapest exactly at $p = q = 5$.

Removing the pendant leaves altogether, so that every w_i has degree q, destroys the effect over the same range: for all 36 pairs with $2 \le p, q \le 7$ the root lies inside $\mathrm{spec}(T)$, with the diagnostic separated by three orders of magnitude (`code/hall_family.py`). The test throughout is the scaling of the density of states at the root, which vanishes linearly in η outside the spectrum and tends to a positive limit inside, and which avoids the cavity mass gate that these flat-band graphs fail.

That observation is real but it does not mean what it appears to mean. Neither minimum degree at least two nor bounded maximum degree repairs Conjecture 10, and a single graph on 31 vertices, smaller than Hall's, refutes both at once.

Proposition 14. *There is a simple connected graph on* 31 *vertices, with every degree equal to two or three, having a root of its matching polynomial in an internal gap of* $\mathrm{spec}(T)$.

Take the rooted binary tree of depth three and identify each of its eight leaves with a vertex of an attached triangle. Internal skeleton vertices have degree three, each attachment vertex has degree three, the remaining triangle vertices have degree two, and the skeleton root has degree two, so the degree set is exactly $\{2, 3\}$ and there are 31 vertices and 38 edges. The matching polynomial has the factor $x^4 - 7x^2 + 8$, whose smaller positive root

$$\theta = \sqrt{\tfrac{7 - \sqrt{17}}{2}} = 1.19935282014558573\ldots$$

lies in an internal gap of spec(T): the gap is narrow, of width about 0.016, with spectrum on both sides. The factorisation was computed twice by independent routes, the rooted pair recursion for a tree of blocks and a memoised vertex-deletion recursion, and the spectral exclusion is certified as in §5: Newton at 40 digits, seeded from the cavity fixed point, gives a real ratio system on all 76 directed edges with residual $5 \cdot 10^{-41}$ and every ratio nonzero, with decay rate

$$\rho = 0.99273509760803033\ldots < 1.$$

The margin $1-\rho = 0.0073$ is thinner than elsewhere in this section but is resolved far beyond the precision used. Verified in `code/subcubic.py`.

The polynomial half is machine-checked in `RamaLean/SubcubicCounterexample.lean`. The quartic is not an accident of the depth: at depth two the recursion already factors as

$$X \cdot A_1 - 2B_1 = X^2(X^2-3)(X^4-7X^2+8)$$

(`quartic_factor`), so the quartic divides A_2 and hence $\mu_G = A_3$ (`quartic_dvd_A3`), whose degree is confirmed to be 31 (`degree_A3`). What depth three supplies is not the root but the gap: at depth two the same root lies inside spec(T), and only the deeper skeleton opens a gap around it. That is worth stating because it separates the two halves of a counterexample cleanly, and says which one is the binding constraint here.

The construction explains why the earlier families needed large degrees: there the number of branches was the degree of the central vertex, so many branches forced high degree. A tree skeleton separates the two, supplying arbitrarily many branches at degree three. What the mechanism needs is branches, not degree.

Proposition 15. *There are simple connected bipartite graphs of minimum degree two with a root of the matching polynomial in an internal gap of* spec(T)*.*

Replace each pendant leaf of $G(p,q)$ by a pendant cycle C_k attached at w_i. Every new vertex then has degree two, as do the middle vertices of $K_{2,q}$, so the graph has minimum degree two, and the branches still meet only at c, so the deletion recurrence still computes μ_G. The effect survives, at a price in size: it needs larger p and q than the pendant-leaf version, which is why a search stopping at $p, q \leq 7$ sees it only for $k = 5$. Counterexamples occur at

$$(k,p,q) \in \{(5,7,7), (4,7,8), (3,8,8), (5,7,8), (5,8,7), (4,8,8), (5,8,8)\},$$

the smallest having 92 vertices, against 41 for Hall's. So the leaves make the construction cheaper without being necessary to it.

For $(k,p,q) = (5,7,7)$, on 92 vertices with 140 edges and degrees between two and nine, the root is the algebraic number $\theta = 2.852901286261237\ldots$, the second largest root of the irreducible octic $x^8 - 26x^6 + 187x^4 - 350x^2 + 91$, which divides $x\mu_H - p\mu_{H-v}$ and hence μ_G.

The spectral exclusion is certified, in a different form from [12] and for a structural reason. A ratio system exists for *every* λ outside spec(A_T), so the ratio equations impose no condition on λ and there is nothing to eliminate; exact algebra in a fixed number field is therefore not available in the way it is for Hall's example, where the ratios happen to lie in $\mathbb{Q}(\sqrt{5}, \sqrt{41})$. Instead the certificate is a posteriori. The 280 directed edges fall into 11 orbits, listed in `code/mindeg2_exact.py` together with their non-backtracking follower counts, and the orbit equations are checked against the full 280-edge fixed point. Newton at 80 digits gives ratios with residual $5.6 \cdot 10^{-81}$, all nonzero; since $\|J^{-1}\|_\infty = 204.8$, the Newton–Kantorovich bound places an exact solution within $1.1 \cdot 10^{-78}$ of the computed one. The decay rate there is

$$\rho = 0.9666941618858803717\ldots, \qquad 1 - \rho = 0.0333,$$

while the entrywise Lipschitz constant of the orbit quotient in the ratios is 45, so ρ can move by at most $5.7 \cdot 10^{-76}$ over the certified ball: the margin exceeds that by a factor $6 \cdot 10^{73}$. Hence $\rho < 1$ for the exact solution, and by [13] $\theta \notin \mathrm{spec}(A_T)$. A Collatz–Wielandt vector gives the same conclusion independently, at 0.9829. The statement is thus rigorous modulo Newton–Kantorovich exactly as it is rigorous modulo [13], both classical. Verified in `code/mindeg2.py` and `code/mindeg2_exact.py`.

5.4 Approximate localization

The counterexamples miss by very little. Locating the gap edges sharply, by bisecting the Angel–Friedman–Hoory decay rate to $\rho = 1$ rather than by scanning a density of states, gives for each of them the distance from the root to $\mathrm{spec}(T)$:

family	n	root	defect
pendant leaf $(7,3)$	43	1.88040	0.0072
pendant leaf $(5,5)$, Hall's	41	2.23607	0.0118
pendant leaf $(7,6)$	64	2.47528	0.0348
pendant C_5 $(7,7)$	92	2.85290	0.0014
pendant C_3 $(8,8)$	97	3.01489	0.0021
triangles, depth 3	31	1.19935	0.00024

Over thirteen counterexamples on 31 to 97 vertices, drawn from two unrelated families, the defect never exceeds 0.035 and shows no growth in n. Every violated gap has width at most 0.22. This suggests that Conjecture 10 is wrong in its constant rather than in its shape.

Conjecture 16 (D1). *There is an absolute constant C such that for every finite graph G and every root θ of μ_G, $\mathrm{dist}(\theta, \mathrm{spec}(T_G)) \le C$.*

The statement was fixed before the confirming data was produced and has survived an out-of-sample test on ten counterexamples disjoint from the three that suggested it. The scaled form with $C/|V(G)|$ is not refuted but is not supported either: defect $\cdot n$ varies by a factor of three hundred with no pattern.

What D1 would buy is machine-checked in `RamaLean/DefectLocalization.lean`. It says exactly that the roots lie in the closed C-neighbourhood of $\mathrm{spec}(T)$ (`zeros_subset_thickening`), so at $C = 0$ it is Conjecture 10 itself (`thickening_zero`); and it restores that conjecture away from the edges:

Proposition 17. *If every root lies within C of $\mathrm{spec}(T)$ and (a, b) is a gap, then no root lies in $(a + C,\ b - C)$.*

So under D1 every gap wider than $2C$ has a root-free core, and any counterexample is confined to a narrow gap, which is what the measured widths show. The open interval is not an artefact: a root may sit exactly at distance C from the spectrum, and the gap edges themselves belong to $\mathrm{spec}(T)$ (`root_free_core`, `conj10_of_zero_defect`).

This is the form in which the question seems worth attacking. It is also the natural one: a root descending to a spectral edge but never passing it is the Alon–Boppana picture, and Conjecture 10 was the assertion that matching roots are Ramanujan in that sense without exception. D1 says the exceptions exist but are uniformly small.

5.5 The mechanism is a separation

Every counterexample is built on a separation. If a vertex v separates G into p isomorphic branches H, the deletion recurrence gives

$$\mu_G = \mu_H^{p-1}\big(x\mu_H - p\,\mu_{H-v}\big), \tag{8}$$

and the root is placed by tuning p and H; this is `CutVertexMechanism.branch_factor`, with `root_of_branch_factor` transferring a root of the last factor to μ_G. Hall's graph and the graphs of §5.3 are of this shape.

A cut vertex is only the smallest separation. Two hubs u, v with p identical branches attached to both leave no cut vertex, since removing either hub leaves the graph joined through the other, while $\{u, v\}$ separates it into p pieces. Writing $A = \mu_B$ for the branch, A_1, A_2 for the branch with one anchor deleted and A_{12} for both, deletion at u and then at v gives

$$\mu_G = A^{p-2}\big(x^2A^2 - px(A_1 + A_2)A + pA_{12}A + p(p-1)A_1A_2\big), \tag{9}$$

which is `SeparationOrder.two_cut_factor`, with `root_of_two_cut`.

The bracket in (9) is *quadratic* in p where the bracket in (8) is linear. Read as polynomials in the branch count the two have degrees one and two, with leading coefficient A_1A_2 in the second case (`cutBracket_natDegree`, `twoCutBracket_natDegree`, `freedom_grows`). Excluding cut vertices does not remove the engine; it upgrades it.

Proposition 18. *There are 2-connected counterexamples to Conjecture 10.*

Let $B_{q,T}$ be $K_{2,q}$ with a path of T further vertices appended to one of its two degree-q vertices; attach the other degree-q vertex to u and the end of the path to v, and take p copies.

branch	p	n	$\kappa(G)$	θ	defect	gap width
$B_{4,2}$	7	58	2	2.137920	0.003574	0.14758
$B_{5,2}$	6	56	2	2.287307	0.005624	0.06636
$B_{5,3}$	6	62	2	2.317609	0.008332	0.09876

All three are simple, bipartite, of minimum degree two and maximum degree at most seven, and of vertex connectivity exactly two, computed by Menger as a unit-capacity maximum flow over every non-adjacent pair. Each θ is an algebraic number of degree 12 or 14 isolated from the exact integer bracket, and (9) is checked against a brute-force matching polynomial. That $\theta \notin \mathrm{spec}(T)$ is confirmed twice: the Angel–Friedman–Hoory decay rate at 40 digits is 0.943 to 0.959, and the density of states of the universal cover, from the complex cavity equations, falls by a factor of ten per decade of η at θ while an in-band control holds constant to four figures (`code/twocut.py`).

Equation (9) is the case $k = 2$ of a general expansion. With k hubs and p branches attached to all of them,

$$\mu_G = \sum_{S \subseteq [k]} (-1)^{|S|} x^{\,k-|S|} \sum_{P \vdash S} (p)_{|P|} \Big(\prod_{J \in P} A_J \Big) A^{\,p-|P|}, \tag{10}$$

with A_J equal to μ_B with the anchors indexed by J deleted and $(p)_m$ the falling factorial: A^{p-k} times a bracket of degree k in p. A k-cut thus needs only k-connectedness while supplying a degree-k polynomial to tune. No counterexample above connectivity two is known: a search over thirty-eight branch families at $k = 3, 4, 5$ with up to 62 vertices, with (10) checked against brute force at $k = 3$, produced none, and it is not vacuous, those graphs carrying two to five gaps of total width 0.40 to 0.75 (`code/kcut.py`).

5.6 The threshold is the minimum degree

Connectivity is not the right invariant, because a graph of vertex connectivity κ has minimum degree at least κ, and it is the minimum degree that the counterexamples respond to. Every counterexample above has minimum degree at most two: Hall's has leaves, the 31-vertex graph of Proposition 15 has degree-2 vertices, and so do all three graphs of Proposition 18.

The two-hub construction separates the two conditions exactly, since it can be run with branches whose every vertex has degree at least three: ladders anchored at both ends of both rails, prisms, and $K_{3,q}$ blocks with two of the three degree-q vertices anchored. That yields 102 graphs of minimum degree three whose vertex connectivity is *two*, so each retains the separating pair the engine needs, and 77 of them have a gap of width at least 0.05, several wider than 2.9, against the 0.066 to 0.148 that the counterexamples of Proposition 18 occupy. None has a root of the bracket in a gap (`code/mindeg3.py`); that sweep did not test the roots of A, which is where the counterexample below lives.

Conjecture 19 (D3)**.** *Every finite graph of minimum degree at least three satisfies* $\mathrm{Zeros}(\mu_G) \subseteq \mathrm{spec}(T_G)$*.*

Conjecture 19 is false. We record the counterexample, which is smaller than Hall's by a factor of three.

Proposition 20 (D3 fails on 14 vertices)**.** *Let G have two hubs u, v, three copies of the star $K_{1,3}$ with centre c_i and leaves $\ell_{i,1}, \ell_{i,2}, \ell_{i,3}$, and every leaf joined to both hubs. Then G is simple, connected and bipartite on* 14 *vertices and* 27 *edges, with* $\deg c_i = \deg \ell_{i,j} = 3$ *and* $\deg u = \deg v = 9$*, so* $\delta(G) = 3$*; and*

$$\mu_G = x^4\,(x^2-3)\,(x^8 - 24x^6 + 171x^4 - 396x^2 + 270),$$

so $\sqrt{3}$ is a root of μ_G, while $\sqrt{3} \notin \mathrm{spec}(T_G)$*.*

The two halves of Proposition 20 are certified to different standards, and we label them rather than average them. That μ_G factors as displayed, and hence that $\sqrt{3}$ is a root, is exact: μ_G is computed from the graph by the deletion recursion and $(x^2 - 3)$ is divided out over $\mathbb{Q}$. That $\sqrt{3} \notin \mathrm{spec}(T_G)$ rests on the density-of-states ladder described below, which is numerical, validated against the covers whose spectrum is known in closed form, and carries an internal control on this same graph; it is not machine-checked. An exact certificate is owed, and the obstruction is stated below: the ratio system, which is what an exact certificate would use, degenerates at precisely this point.

The mechanism is Hall's verbatim. His violating root $\sqrt{5}$ is the top matching root of the star $K_{1,5}$, whose matching polynomial is $x^4(x^2-5)$; the divisor is a star. A star's leaves have degree one *in the divisor* however large their degree is *in* G, so a hypothesis on $\delta(G)$ constrains the ambient graph and not the divisor. Here the branch is $K_{1,3}$, $A = \mu_{K_{1,3}} = x^2(x^2-3)$, and the two-cut identity $\mu_G = A^{p-2}\big(x^2A^2 - px(B_u+B_v)A + pDA + p(p-1)B_uB_v\big)$ makes every root of A a root of μ_G once $p \geq 3$. Three branches are needed and two do not suffice, since $A^0 = 1$ carries no root.

This is exactly what the searches above could not see, and the reason is worth stating because it is not a matter of scale. Each of them lifted the block's degrees to three by adding edges *inside* the block, which changes A and moves its roots; the counterexample lifts them from *outside*, by attaching each leaf to both hubs, which leaves A untouched. The five clean searches were therefore all run on blocks whose matching polynomial had already been perturbed away from the star. Separately, `code/mindeg3.py` iterated over the roots of the bracket and never over the roots of A, so the divisibility that is the whole engine went untested; that is repaired in `code/mindeg3adv.py`, which reports a signed margin rather than a verdict.

Certifying $\sqrt{3} \notin \operatorname{spec}(T_G)$ needs the density-of-states ladder and not the ratio system, and the reason is instructive. The Angel–Friedman–Hoory system $\lambda = 1/r_e + \sum_{e\to f} r_f$ has four orbits here under $\operatorname{Aut}(G)$, and eliminating gives a quadratic whose two real branches both have decay above one, which reads as $\sqrt{3} \in \operatorname{spec}(T_G)$. That is wrong: a hair either side of $\sqrt{3}$ there are *three* real branches and the third has decay about 0.87. Its $r(\ell \to u)$ crosses zero exactly at $\sqrt{3}$, so the elimination multiplies by it and drops the branch, and a pointwise solver returns nothing there for the same reason. The ratio parametrisation has a coordinate singularity at the point in question. The resolvent has none: with $g_e = 1/(z - \sum_{e\to f} g_f)$ at $z = \lambda + i\eta$, the quantity $|\operatorname{Im}\sum_w G_w|$ decays linearly in η outside the spectrum, tends to a positive constant inside a band, and diverges like η^{-1} at an atom. The distinction between the first and the third is not a formality here, since $\operatorname{spec}(T)$ is closed and the biregular cover already carries an isolated $\{0\}$; showing every point near $\sqrt{3}$ is outside would not show $\sqrt{3}$ is. On this graph the ladder reports an atom at 0, a band at 1.68, a band at 2.40, and linear decay at $\sqrt{3}$, the atom at the origin serving as the internal control (`code/d3_counterexample.py`).

What survives is the connectivity statement, which Proposition 20 does not touch: G there has vertex connectivity two, the two hubs being a separating pair. So the implication D3 $\Rightarrow$ C2 (`MinimumDegreeThreshold.C2_of_D3`) and the sharpness statement (`threshold_is_three`) remain valid as implications but are now vacuous, their hypothesis being false, and the 3-connected statement C2 stands on its own rather than as a corollary.

The evidence that made Conjecture 19 plausible is recorded below rather than deleted, because a false conjecture that survives five independent searches is worth a note on how it managed it. First, the regular case is trivial and worth separating off: for d-regular G the universal cover is the d-regular tree, whose spectrum is exactly the interval $[-2\sqrt{d-1}, 2\sqrt{d-1}]$ in which Heilmann and Lieb confine every root, so there is nothing to violate (`conj10_of_regular`). The content lies in irregular graphs.

Second, minimum degree three narrows the gaps but does not remove them, so Conjecture 19 is not true for want of anywhere to hide: over 38 graphs the mean total gap width falls from 0.99 at $\delta \le 2$ to 0.09 at $\delta \ge 3$, while individual gaps at $\delta = 3$ reach 2.9 (`code/gapscale.py`).

Third, the evidence covers a family unlike the rest. Wheels, a hub joined to a cycle, have minimum degree three, are non-bipartite by construction, and carry gaps from $(2.02, 2.08)$ at W_5 out to $(2.02, 3.20)$ at W_{11}. None up to W_{14} has a root in any gap, and at the closest approach the verdict rests on the density-of-states ladder rather than on locating a gap edge: for W_{10} through W_{20} the largest root below 2 has density of states constant to four significant figures across four decades of η, so it lies in the spectrum (`code/wheels.py`).

Fourth, and this is what the evidence amounted to. Five searches were run against Conjecture 19, each designed against the failure mode of the last, and all came back clean; Proposition 20 shows they were all clean for one reason, that each raised the block's degrees by adding edges inside it. Two reproduce Hall's own mechanism at minimum degree three — p copies of a block glued at a new centre, so that the branch factorization makes every root of μ_H a root of μ_G — first with off-the-shelf blocks and then with blocks built to the shape of his, $K_{2,q}$ with a graph on the q-side lifting those degrees to three. A third abandons the construction entirely: random graphs of minimum degree three with deliberately spread degrees and *no cut vertex*, which is the case a cut-based engine structurally cannot reach. A fourth requires a separating pair. A fifth holds the degree sequence fixed, so the gap does not move, and varies only the block's internal structure, so the roots do.

engine	*scale*	*outcome*
1-cut, off-the-shelf blocks	118	closest approach 0.169
1-cut, blocks of Hall's shape	350	clean
separator-free random search	419	clean
separating-pair random search	39	clean
fixed degrees, varying structure	14	clean

About twelve hundred configurations, and the two closest approaches are both cut-based (`code/D1cut.py`, `code/D1cut_adv.py`, `code/D3broad.py`, `code/D3fixdeg.py`). The nearest miss is worth stating because it is nearer than the table suggests: a block $K_{2,12}$ with a perfect matching on the 12-side has $\mu_H = (x-1)^4(x+1)^4(x^6 - 26x^4 + 157x^2 - 12)$, so 1 is a root of multiplicity four and hence a root of μ_G, and 1 lies well inside the central gap of half-width $\sqrt{11} - \sqrt{2} = 1.90$. It is not a counterexample only because 1 is a spectral *atom* of the cover, so it lies in $\mathrm{spec}(T)$ after all: the density of states there reads 8.97, 89.7 and 897 as η falls through 10^{-2}, 10^{-3}, 10^{-4}, the signature of a point mass. Breaking the symmetry that creates the atom moves the root off it — and into the continuous spectrum, not into the gap, since the same change moves the degrees that place the gap.

The separator-free class is not merely clean but comfortable, which is a different statement and the one worth having. Taking the density of states at a root as the margin, since a violation is exactly a zero there, the median over 419 graphs is stable at 0.15 from $n = 10$ to $n = 20$ and if anything rises; the *minimum* falls like $1/n$, but the number of roots sampled doubles over that range and a minimum over more draws is smaller for that reason alone, the observed 4.19×10^{-2} against the 4.04×10^{-2} that sample size predicts (`code/D3margin.py`). So Conjecture 19 is not surviving these tests by a hair.

There is one reformulation, and it comes from the same measurements rather than from a new idea. Godsil gives $\mu_G \mid \mu_P$ for the path tree P, and P is a tree, where the matching polynomial is the characteristic polynomial; so every root of μ_G is an eigenvalue of P, and $\mathrm{spec}(P) \subseteq \mathrm{spec}(T)$ implies the conjecture (`PathTreeRoute.conj_of_pathtree`). The hypothesis is strictly stronger — μ_P carries its own roots too — and it fails for Hall's graph exactly where the conjecture does. The containment holds for the biregular families measured, but at minimum degree three it is *false*: one graph in 86 carrying a gap has a path tree with twelve eigenvalues inside it while no root of μ_G is there. So the hypothesis fails exactly where Conjecture 19 holds, and the reduction cannot carry a proof of it; the companion paper [16] gives the counterexample. We record it because it closes a route that the biregular evidence makes look plausible.

What none of it supplies is an argument. Each engine was built against the previous failure mode and each came back clean, which raises confidence and yields nothing proof-shaped. Every counterexample known has a separation, and the evidence is strongest exactly where separations are absent; turning that observation into a theorem is the open problem, and it is not one more search.

5.7 A second conjecture falls with it

The counterexample refutes more than the conjecture it was built for. Zili Xu [17] proposes a spectral-inclusion conjecture for *additive products* in the sense of Mohanty–O'Donnell [18]: with $A_1, \ldots, A_c$ adjacency matrices on a common vertex set, $X = \mathrm{AddProd}(A_1, \ldots, A_c)$, and

$$\alpha(A_1, \ldots, A_c; x) = \mathbb{E} \det\Big(xI - \sum_j Q_j^* A_j Q_j\Big)$$

the additive characteristic polynomial, the expectation over independent uniform ± 1 diagonal signings, he conjectures $Z(\alpha) \subseteq \mathrm{Spec}(X)$. Mohanty–O'Donnell prove the weaker $Z(\alpha) \subseteq [-\rho(X), \rho(X)]$, so the proposal stands to that theorem exactly as Conjecture 10 stands to Hall–Puder–Sawin.

His own remark [17] supplies the refutation. Taking the atoms to be the individual edge-adjacency matrices of a finite graph G gives $\alpha = \mu_G$ and $\mathrm{AddProd} = \mathrm{UCT}(G) = T$, so the conjecture specialises to $Z(\mu_G) \subseteq \mathrm{Spec}(T)$, which is Conjecture 10 at $d = 1$. Hall's graph therefore refutes it, and the accompanying (k, m)-characteristic-polynomial restatement falls with it through the identity $\psi_{k,m}[k\mathcal{A}_{\mathrm{bd}}] = \alpha$.

Proposition 21. *The additive-product spectral-inclusion conjecture is false, as is its (k, m) restatement.*

Proof. Instantiate at the edge-atom system of the graph of Theorem 11. By the edge-atom identification $\alpha = \mu_G$, so $\alpha(\sqrt{5}) = 0$ by (7), while $\sqrt{5} \notin \mathrm{Spec}(T)$ by the certificate of §5. The (k, m) form follows by substituting the identity above. □

The assembly is machine-checked in `RamaLean/XuAdditiveProduct.lean` as `xu_conj21_false` and `xu_conj24_false`, with the edge-atom identification and Angel–Friedman–Hoory entering as the hypotheses they are; the root and the exclusion come from the verified `muG_root_sqrt5` and the certificate of Parts B–D.

There is a dichotomy attached to that proposal which our answer sharpens. Either the conjecture holds for all additive products, in which case the graph statement follows by the edge-atom specialisation; or it fails somewhere, in which case real-rootedness, the free-like walk moment identities and the spectral-radius bound are together insufficient for spectral inclusion, and any proof of the graph statement must use a further property of edge-atom systems. The second branch holds, and more sharply than that: the failure is already *in* the edge-atom case (`xu_dichotomy_second_branch`), so no additional property of edge-atom systems can rescue the statement, and the repair must restrict the graph class. That is what §5.6 does.

5.8 What survives

The counterexample has $b_1 = 20$, feedback vertex number 5, and is neither biregular nor of bounded feedback vertex number, so it violates the hypotheses of every positive result in this paper. Theorem 1 and the closed form at $b_1(G) = 1$, the subdivision case, the complete bipartite case of the companion paper [16], the reductions of §4.3, the localization $\mathrm{Zeros}(\mu_G) \subseteq \mathrm{spec}(G^{\mathrm{ab}})$ and the sufficient conditions of the companion paper [16] are all unaffected. What changes is the question. The counterexample lives exactly in the residue $\mathrm{spec}(G^{\mathrm{ab}}) \setminus \mathrm{spec}(T)$ that §4.3 identifies as the only place a counterexample could live, so the framework was pointed at the right region; what it lacked was any reason for that region to be empty. The open problem is to characterise the graphs for which $\mathrm{Zeros}(\mu_{d,G}) \subseteq \mathrm{spec}(T)$ holds; Conjecture 19 is our proposal, and §5.6 records its evidence and its sharpness. The content lies entirely in graphs whose universal cover has gaps, and the canonical such family is the (a, b)-biregular one, which is Problem 1 of Song, Fan and Miao.

6 Related work and formalization

The d-matching polynomial and the interval $[-\rho, \rho]$ are due to Hall, Puder and Sawin; the counterexample refuting the inclusion is due to Chris Hall [12] and is reproduced here with his permission, with our own independent verification of its certificate. The Angel–Friedman–Hoory ratio criterion [13] is what certifies the gap.

The Lean 4/Mathlib sources are in `RamaLean/` of the accompanying repository, and carry no `sorry` and no custom axioms. Formalized here: the closed form at first Betti number one and its corollaries, the counterexample's algebra, the branch and two-cut factorizations, the separation mechanism, and the minimum-degree threshold, including that minimum degree two does not suffice and that three is therefore exact. Statements are labelled `VERIFIED`, `HEURISTIC` or `CONJECTURE` in the sources, and the labels are used with their plain meaning: a `HEURISTIC` claim is supported by computation and is not proved.

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